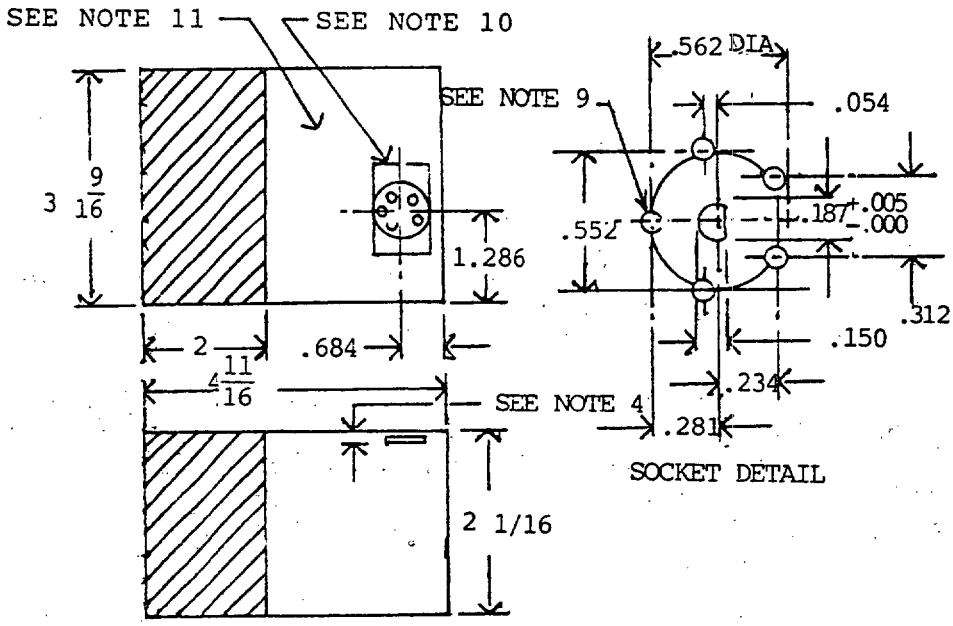
MILITARY SPECIFICATION SHEET

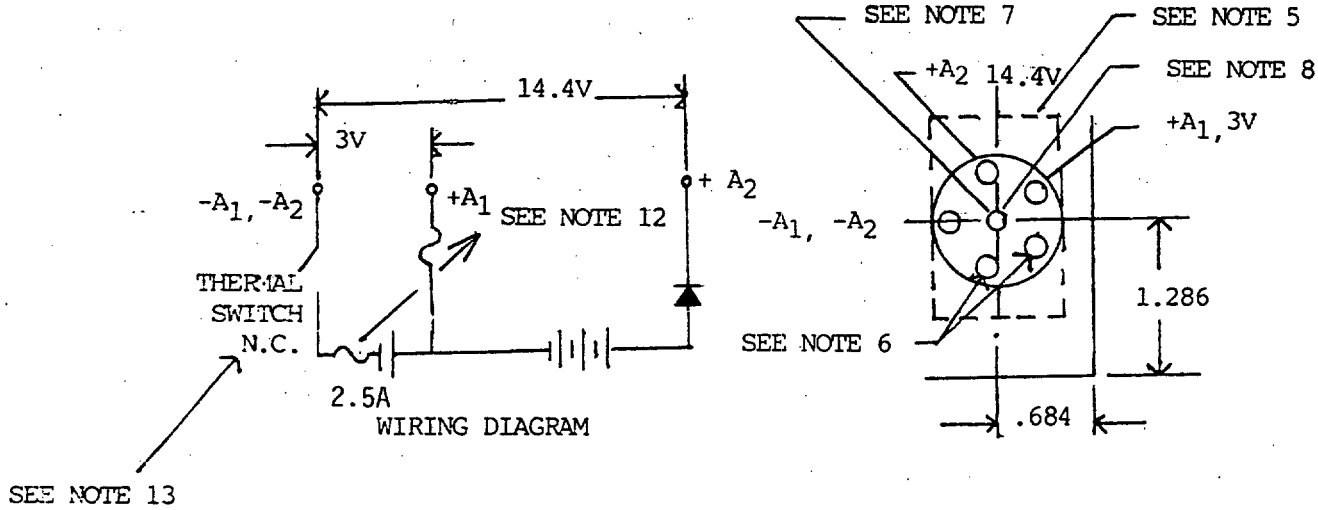
BATTERY, NON-RECHARGEABLE, LITHIUM THIONYL CHLORIDE, BA-6598/U

This specification is approved for use by Laboratory Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the batteries described herein shall consist of this document and the latest issue of MIL-B-49461(ER).

INCHES	MM
0.005	0.13
0.054	1.37
0.150	3.81
0.187	4.75
0.234	5.94
0.281	7.14
0.312	7.92
0.552	14.02
0.562	14.27
0.662	16.81
0.684	17.37
1.286	32.66
2.000	50.80
2-1/16	52.39
3-9/16	90.49
4-11/16	119.06





AMSC N/A FSC 6135

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NOTES

1. All dimensions are in inches.
2. Unless otherwise specified, all tolerances are ±0.005 inch for decimal.
3. Tolerances of length, width, and height of battery are ±1/16 inch for fractions.
4. The minimum depth of socket well shall be 1/2 inch.
5. The size of the insulating plate is optional.
6. Socket insert contact not required.
7. Pin-circle center.
8. "D" hole for plug positioning.
9. Five holes to fit 1/8 inch in length.
10. Socket must float 3/32, +1/32,-1/16 inch in all directions from the specified pin circle center location in the plane of the socket from reference dimension.
11. The socket shall be supported and mounted so that the top surface of the socket shall not protrude above the adjacent outside surface of the jacket and shall not be more than 1/16 inch below adjacent outside surface of the jacket when used initially, during, and after subsequent insertions of the mating plug during contractual testing.
12. A 2.5 ampere slow blow non-replaceable fuse shall be used in the negative leg and the $+A_1$ leg.
13. The thermal switch shall be physically located as near as possible to the central most cell. It may be wired at any point in the electrical circuit of the battery.
14. The battery shall have a diode adjacent to the slow blow fuse and in series with it to prohibit charging.

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REQUIREMENTS

Dimensions and configuration: See figure 1.

	A_1 Unit (volts)	Load (ohms)	A_2 Unit (volts)	Load (ohms)
Maximum Voltage Under Indicated Load:	3.05	6.05	15.25	14.20
Minimum Voltage (for capacity):	2.00		10.00	

Terminals: Five (5) hole socket type with no obstruction of any holes.

Weight (maximum): 1.7 pounds (772 grams)

Capacity tests: When the battery is tested in accordance with the methods of examinations and tests of this specification, the minimum capacity test requirements shall be as specified below:

Capacity Test	Battery Capacity in hours to 10.00 Volts
I	7.0
L	3.5
H	13.0
HT	12.6
LT	3.0
T	7.0
D	7.0
ILR	1000

Initial Voltage Delay: When the battery is subjected to the I, H, HT, T, D and ILR capacity test, the time required at the beginning of discharge of the battery to reach a voltage of 10.00 volts after the specified load is applied shall not be more than 5.0 seconds. For the L and LT capacity tests the time to reach a voltage of 10.00 volts after the load is applied shall not be more than 60 seconds.

High temperature switch: A high temperature switch or thermal fuse shall be employed internal to the battery in the negative leg.

METHODS OF EXAMINATION AND TESTS:

(1) Storage: Details on storage conditions for all capacity tests are specified in basic specification.

(2) Discharge:

Cells: Discharge at 2.0 amperes constant current at -20 °F for a minimum of 3.5 hours to 2.50 volts cut-off. Discharge to zero volts. (see paragraph 4.8.13.1).

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Batteries: The A_2 unit ($-A_1$ $-A_2$ to $+A_2$) shall be discharged as follows:

For all capacity tests except the H, HT, and ILR, the batteries shall be discharged at two amperes constant current to zero volts. For the H and HT capacity tests the batteries shall be discharged through two amperes constant current for five minutes followed by open circuit for five minutes. This cycle shall be repeated until the closed circuit voltage reaches zero volts. For the I capacity test the batteries shall be force discharged at one ampere constant current for 7.2 hours immediately after zero volts is obtained. There shall be no venting, fire, or explosion when the batteries are force discharged. For the ILR test a fixed resistance of 1100 ohms shall be used to cut-off voltage of 10.0 volts or 1000 hours.

Closed circuit voltage: Closed circuit voltage measurements shall be made with a direct current voltmeter of proper range and sensitivity (see basic specification) using load resistances as shown in the following table:

	UNIT RESISTANCE	MINIMUM PERMISSIBLE VOLTAGE
A_1	6.5 ohms, $\pm$1%	2.12 volts
A_2	14.2 ohms, $\pm$1%	12.0 volts

Voltage shall be above "MINIMUM PERMISSIBLE VOLTAGE" within five (5) seconds after load is applied.

Socket strength test (First Article only): Before conducting the jacket integrity and visual mechanical examinations specified in Table III, the 5 batteries shall be subjected to the following:

Using a 0.250" $\pm$ 0.050" diameter rod, apply a force to the center of the socket and deflect it downward a distance of 0.250" + 0.0625", -0" below the top surface of the battery.

Custodian:
Army-ER

Preparing Activity:
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(Project 6135-A203)

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